FUNCTORIALITY OF THE SHEAF OF DIFFERENTIALS FOR A COMMUTATIVE DIAGRAM OF RINGED TOPOI

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Both the statement and the proof contain commutative diagrams typeset with Xy-pic; the retained two-slice preamble of this source supplies the native Xy-pic setup, so they compile as in the source.

Editorial note. The counted span cites no labelled external item, so no navigation stubs were needed.

Allowed standard prerequisites (editorial).

- Sheaves of differentials on ringed topoi and their universal derivation. Stacks Project, Modules on Sites, section on differentials (definition of Omega and d_{X/Y}). Supplies the objects and the universal derivation whose functoriality is being established.
- Adjointness of f_* and f^* for a morphism of ringed topoi. Stacks Project, Modules on Sites, section on morphisms of ringed topoi. Converts the map into the pushforward form into the map c_f in the pullback form.

Lemma 0.1. *Let $X = (Sh(\mathcal{C}_X), \mathcal{O}_X)$, $Y = (Sh(\mathcal{C}_Y), \mathcal{O}_Y)$, $X' = (Sh(\mathcal{C}_{X'}), \mathcal{O}_{X'})$, and $Y' = (Sh(\mathcal{C}_{Y'}), \mathcal{O}_{Y'})$ be ringed topoi. Let*

$$\begin{array}{ccc} X' & \xrightarrow{f} & X \\ \downarrow & & \downarrow \\ Y' & \longrightarrow & Y \end{array}$$

be a commutative diagram of morphisms of ringed topoi. The map $f^\sharp : \mathcal{O}_X \to f_*\mathcal{O}_{X'}$ composed with the map $f_* d_{X'/Y'} : f_*\mathcal{O}_{X'} \to f_*\Omega_{X'/Y'}$ is a Y-derivation. Hence we obtain a canonical map of $\mathcal{O}_X$-modules $\Omega_{X/Y} \to f_*\Omega_{X'/Y'}$, and by adjointness of f_* and f^* a canonical $\mathcal{O}_{X'}$-module homomorphism

$$c_f : f^*\Omega_{X/Y} \longrightarrow \Omega_{X'/Y'}.$$

It is uniquely characterized by the property that $f^* d_{X/Y}(t)$ maps to $d_{X'/Y'}(f^*t)$ for any local section t of $\mathcal{O}_X$.

Proof. This is clear except for the last assertion. Let us explain the meaning of this. Let $U \in \mathrm{Ob}(\mathcal{C}_X)$ and let $t \in \mathcal{O}_X(U)$. This is what it means for t to be a local section of $\mathcal{O}_X$. Now, we may think of t as a map of sheaves of sets $t : h_U^\# \to \mathcal{O}_X$. Then $f^{-1}t : f^{-1}h_U^\# \to f^{-1}\mathcal{O}_X$. By f^*t we mean the composition

$$f^{-1}h_U^\# \xrightarrow{f^{-1}t} f^{-1}\mathcal{O}_X \xrightarrow{f^\sharp} \mathcal{O}_{X'}, \qquad f^{-1}h_U^\# \xrightarrow{f^*t} \mathcal{O}_{X'}$$

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Note that $\mathrm{d}_{X/Y}(t) \in \Omega_{X/Y}(U)$. Hence we may think of $\mathrm{d}_{X/Y}(t)$ as a map $\mathrm{d}_{X/Y}(t) : h_U^\# \to \Omega_{X/Y}$. Then $f^{-1}\mathrm{d}_{X/Y}(t) : f^{-1}h_U^\# \to f^{-1}\Omega_{X/Y}$. By $f^*\mathrm{d}_{X/Y}(t)$ we mean the composition

$$f^{-1}h_U^\# \xrightarrow{f^{-1}\mathrm{d}_{X/Y}(t)} f^{-1}\Omega_{X/Y} \xrightarrow{1\otimes\mathrm{id}} f^*\Omega_{X/Y}, \qquad f^{-1}h_U^\# \xrightarrow{f^*\mathrm{d}_{X/Y}(t)} f^*\Omega_{X/Y}$$

OK, and now the statement of the lemma means that we have

$$c_f \circ f^*t = f^*\mathrm{d}_{X/Y}(t)$$

as maps from $f^{-1}h_U^\#$ to $\Omega_{X'/Y'}$. We omit the verification that this property holds for c_f as defined in the lemma. (Hint: The first map $c_f' : \Omega_{X/Y} \to f_*\Omega_{X'/Y'}$ satisfies $c_f'(\mathrm{d}_{X/Y}(t)) = f_*\mathrm{d}_{X'/Y'}(f^\sharp(t))$ as sections of $f_*\Omega_{X'/Y'}$ over U, and you have to turn this into the equality above by using adjunction.) The reason that this uniquely characterizes c_f is that the images of $f^*\mathrm{d}_{X/Y}(t)$ generate the $\mathcal{O}_{X'}$-module $f^*\Omega_{X/Y}$ simply because the local sections $\mathrm{d}_{X/Y}(t)$ generate the $\mathcal{O}_X$-module $\Omega_{X/Y}$. $\square$